# OpTool User Guide

by Carsten Dominik, Michiel Min, and Ryo Tazaki; September 2023, version 1.9.14

## 1. Introduction

This tool produces complex dust particle opacities right from the command line. It is derived from Michiel Min's DHS OpacityTool and also implements Ryo Tazaki's MMF theory for highly porous aggregates.

### 1.1. Capabilities

- stand-alone tool, fully command line driven, no input files need to be edited
- full scattering matrix output in several formats, including for RADMC-3D
- combining materials through mixing into a complex grain with porosity
- *built-in*: a curated collection of materials for applications in astronomy
- external refractive index data can be used just as easily
- computational methods: (i) **DHS (Distribution of Hollow Spheres)** for *irregular grains* and *low-porosity* aggregates. Standard **Mie theory** for *perfect spheres* is available as a limiting case. (ii) **MMF (Modified Mean Field)** theory for *high-porosity/fractal aggregates*. (iii) **CDE** approximation in the Rayleigh limit.
- `Python` interface module for plotting and post-processing

### 1.2. Terms of use

`optool` is distributed under the MIT license and can be used, changed and redistributed freely. But we do ask you to provide a reference to `optool` when using it. Relevant references are listed below and the corresponding BibTeX entries are available in the file `optool.bib`. `optool` is hosted on github.

- **optool:** Dominik, C., Min, M. & Tazaki, R. 2021, Optool, 1.9, Astrophysics Source Code Library, ascl:2104.010
- **DHS model for irregular grains:** Min, M. et al. 2005, A&A, 432, 909
- **MMF model for aggregates:** Tazaki, R. & Tanaka, H. 2018, ApJ 860, 79
- **DIANA standard opacities:** Woitke, P. et al. 2016, A&A 586, 103
- References to refractive index data used in your particular application.

### 1.3. Physical units used by optool

Due to conventions in our field, the input and output of `optool` uses the following units:

| | |
|---|---|
| grain sizes and wavelengths.[1] | $\mu$m |
| mass densities of materials | g cm$^{-3}$ |
| opacities $\kappa_{\rm abs}$, $\kappa_{\rm sca}$, $\kappa_{\rm ext}$ | cm$^2$ g$^{-1}$ |
| scattering matrix, see App. B.1 | sr$^{-1}$ *or* cm$^2$ g$^{-1}$ sr$^{-1}$ |

[1] When giving a grain size or a wavelength on the command line, you can write `1.3*mm`, `340*GHz`, or `4000/cm` and `optool` will do the right thing, converting to 1300$\mu$m, 881.7$\mu$m, and 2.5$\mu$m, respectively.

## 2. Examples

A simple grain made only of the default pyroxene, for the default grain size distribution ($a^{-3.5}$ powerlaw from 0.05 to 3000$\mu$m), on the default wavelength grid (0.05$\mu$m to 1cm).

```
optool pyr
```

Include the scattering matrix in the produced output

```
optool pyr -s
```

Reproduce the DIANA standard dust model, using a specific pyroxene (70% Mg) and carbon, in a mass ratio 0.87/0.13, and with a porosity of 25%.

```
optool pyr-mg70 0.87  c 0.13  -p 0.25
```

List the built-in materials

```
optool -c
```

Add a water ice mantle (built-in data from Warren+08) that is 20% of the core mass

```
optool pyr-mg70 0.87  c 0.13  -m h2o-w 0.2  -p 0.25
```

Like the previous example, but use ice refractive index data from a separate file.

```
optool pyr-mg70 0.87  c 0.13  -p 0.25  -m data/ice_hudgins.dat 0.2
```

Pure water ice grains in a narrow size distribution from 1 to 3 microns, with 15 sample sizes following an $f(a) \propto a^{-2.5}$ powerlaw size distribution. Also, restrict the wavelength range to 10-100$\mu$m, and turn off DHS to get perfect spheres (Mie).

```
optool h2o  -a 1 3 2.5 15  -l 10 100 -mie
```

Use a log-normal size distribution around 2 $\mu$m with $\sigma$=0.7 instead.

```
optool h2o  -a 0.1 30 2.0:0.7  -l 10 100 -mie
```

For silicon carbide, compute the opacity of a single grain size (2.5$\mu$m) at $\lambda$=8.9$\mu$m.

```
optool -a 2.5 -l 8.9 sic
```

Represent the default dust model (DIANA, you also get this when you do not give any materials at all) in 42 grain sizes, and produce input files for RADMC-3D, one for each grain size, with full scattering matrix, chopping 3 degrees from the scattering peak.

```
optool -na 42 -d -s -radmc -chop 3
```

Use MMF to compute the opacities of dust aggregates made of pyroxene monomers. Use a monomer radius of 0.3 $\mu$m to construct aggregates with compact-volume radii between 10 and 30 $\mu$m, and a fractal dimension of 1.9.

```
optool pyr  -a 10 30  -mmf 0.3 1.9
```

Compute CDE for small graphite grains

```
optool gra  -a 0.01 0.1 -l 1 30 -cde
```

# 3. Installation

You can download, compile, and install `optool` with these simple steps, using the freely available GNU FORTRAN compiler `gfortran`.

```
git clone https://github.com/cdominik/optool.git        # clone repository
cd optool                        # enter code directory
make multi=true                  # compile with multicore support

make install bindir=~/bin/       # optional: copy binaries to binary path
pip install -e .                 # optional: install the python module
```

In the compilation step, use `multi=true` to add multicore support (recommended!), `ifort=true` to use the Intel fortran compiler, and `oldio=true` if your compiler does not have the `ISO_FORTRAN_ENV` module.
The executable is called `optool`, run it with `./optool` or move it onto your binary path.

# 4. Command line arguments

`-h [OPT]`
: Show command line option summary, or specific information about option `-OPT`.

`-q` Reduce output to STDOUT to essential warnings and errors.

`-v` More verbose output to STDOUT.

## 4.1. Grain composition

If no composition is specified, the (DIANA) default is **-c pyr 0.87 -c c 0.13 -p 0.25**.

`-c` List available built-in materials (the keys for the **-c** and **-m** options).

`[-c] MATERIAL [MFRAC]`
: Specify a material to include in the grain. `MATERIAL` can be the key for a builtin material, the path to an `lnk` file, or colon-separated numbers `n:k:rho`[2]. `MFRAC` is the *mass* fraction (default 1.0) of the material. You can give up to 20 materials to build up the grain. Mass fractions do not have to add up to one, they will be renormalized. All materials will be mixed together using the *Bruggeman* rule, and vacuum can be added through the porosity. **-c** in front of each material is optional.

`-m MATERIAL [MFRAC]`
: Like **-c**, but place this material into the grain mantle. Multiple mantle materials will be mixed using the Bruggeman rule, and then that mix will be added to the core using the *Maxwell-Garnett* rule. The **-m** is *not* optional, it must be present.

`-p POROSITY [P_MANTLE]`
: Porosity, the *volume* fraction of vacuum, a number smaller than 1. The default is 0. A single value will apply to both core and mantle, but a second value will be specific for the mantle (and may be 0).

`-diana, -dsharp, -dsharp-no-ice`
: Use DIANA (Woitke+2016) or DSHARP (Birnstiel+2018) compositions.

[2] `n:k:rho` specifies a refractive index `m=n+ik` and a material density `rho` (in g/cm$^3$) for a quick calculation at a single wavelength, for example for use in statistical inference on a refractive index.

## 4.2. Grain geometry and computational method

If no method is explicitly specified, the default is **-dhs 0.8**, i.e. DHS with $f_{max}$=0.8.

`-dhs [FMAX]`
: Use the *Distribution of Hollow Spheres* (DHS, Min+ 2005) approach to model deviations from perfect spherical symmetry and low-porosity aggregates. Spheres with inner holes with volume fractions between 0 and $f_{max}$ (default 0.8) are averaged to mimic irregularities. $f_{max}$=0 means to use solid spheres (Mie theory), i.e. perfectly regular grains. For backward compatibility, **-fmax** can be used instead of **-dhs**.

`-mmf [A0 [DFRAC-OR-FILL [KF]]]`
: Use *Modified Mean Field* theory (MMF, Tazaki & Tanaka 2018) to compute opacities of highly porous or fractal aggregates. **-c**, **-m**, and **-p** determine the composition of monomers with radius `A0` (default 0.1$\mu$m). Particles will be aggregates with a *compact size* given by the **-a** switch, giving rise to $N = a^3/a_0^3$ monomers. `DFRAC-OR-FILL` specifies either the fractal dimension (if >1) or the *volume filling factor* (if <1). The default is 0.2. KF may be used to change the default prefactor.

`-mie`
: Do a standard Mie calculation for perfect spheres. This is short for **-dhs 0**.

`-cde`
: Compute CDE (continuous distribution of ellipsoids) Rayleigh limit opacities.

## 4.3. Grain size distribution

`-a AMIN [AMAX [APOW [NA]]]` **(powerlaw size distribution)**
: Specify (minimum) grain radius, and optionally maximum grain radius, the size distribution powerlaw and the number of size bins. You may also use options to set individual values with **-amin**, **-amax**, **-apow**, **-na**. The defaults are 0.05 $\mu$m, 3000 $\mu$m, 3.5, and *15 per size decade with a fixed minimum of 5*, respectively.
> If only a single size is specified with **-a**, then $a_{max}$=$a_{min}$ and $n_a$=1 are implied.

`-a AMIN AMAX AMEAN:ASIG [NA]` **([log-]normal size distribution)**
: Specify the centroid size and the logarithmic width for a log-normal size distribution. You may also use **-amean** and **-asig** options to set these values. If `ASIG` is negative, create a normal distribution with that width (in $\mu$m) around `AMEAN`.

`-a FILE`
: Read the size distribution from a file. The file format is described in appendix A. To get an example file `optool_sd.dat`, run `optool` with the option **-w**.

## 4.4. Wavelength grid

`-l LMIN [LMAX [NLAM]]`
: Specify the (minimum) wavelength, and optionally the maximum wavelength and the number of wavelengths points for the construction of the wavelength grid. The default values are 0.05 $\mu$m, 10000 $\mu$m, and 300, respectively. You may also use the options **-lmin**, **-lmax**, and **-nlam** (or **-nl**) to set individual values.
> If only one wavelength is specified with **-l**, then $\lambda_{max}$=$\lambda_{min}$ and $n_\lambda$=1 are implied.

`-l FILE`
: Read the wavelength grid from `FILE`. To get an example file `optool_lam.dat`, run `optool` with the option **-w**. An `lnk` file could be used here as well!

## 4.5. Controlling the output

The standard output is the file `dustkappa.dat`, with the opacities and the asymmetry parameter $g$. The following options control and extend the output.

`-o [DIR]`
: Put the output files in directory `DIR` instead of the current working directory. `./output` will be used if **`-o`** is present but `DIR` is not specified.

`-s [NANG]`
: Include the scattering matrix in the output. NANG may optionally change the number of equally-spaced angular grid points to cover the range of angles between 0 and 180 degrees. The default for `NANG` is 180 and should normally be just fine.

`-d [NSUB]`
: Divide the computation up into $n_a$ parts to produce a file for each grain size. Each size will be an average over a range of `NSUB` (default 5) grains around the real size.

`-chop [NDEG]`
: Cap the first `NDEG` (2 if unspecified) degrees of the forward scattering peak.

`-radmc [LABEL]`
: RADMC-3D uses a different angular grid and scattering matrix normalization. File names will contain `LABEL` if specified and have the extension `.inp`.

`-print [KEY]`
: Write to `STDOUT` instead of files. The default is to write $\lambda$, $\kappa_{\rm abs}$, $\kappa_{\rm sca}$, $\kappa_{\rm ext}$, and g. Many other outputs are possible, run `optool -print ?` for a full list. For readability, a header line may be printed to `STDERR`, but `STDOUT` gets only numbers which can be used in pipes and for redirection. You can use this to extract a single value, for example the 850$\mu$m extinction opacity of grains between 1 and 3mm: `optool -a 1000 3000 -l 850 -print kext`.

`-w`
: Write the files `optool_sd.dat` and `optool_lam.dat` with the grain size distribution and the wavelength grid, respectively. Also, write `optool_mix.lnk` with the result of mixing refractive index data. Exit without doing a computation.

# 5. Material properties

`optool` needs refractive index data to work. For your convenience, a useful list of materials is compiled into `optool`. You can also find and use other data.

## 5.1. Built-in materials

To access one of the built-in materials, specify the corresponding key string like `pyr-mg70`. In each material class we have selected a useful default, accessible with an even simpler generic key (for example, `pyr` is an alias for `pyr-mg70`). Most of the built-in refractive index datasets have a reasonably wide wavelength coverage - the few exceptions are highlighted by bold-face numbers. If a material is being used outside of the measured region, `optool` will still function, using extrapolated optical properties.
Even the limited number of materials we have selected to include with `optool` can be daunting. To get started with some kind of standard opacity, we recommend to work with pyroxene [pyr], carbon [c], and, at low temperatures, water ice [h2o] (Woitke+ 2016). If you need to account for sulfur, you may want to include troilite [tro] (Birnstiel+ 2016).

| **-c Key** generic | **-c Key** full key | **Material** | **State** | $\rho$ g/cm$^3$ | $\lambda_{min}$ $\mu$m | $\lambda_{max}$ $\mu$m | **Reference** |
|---|---|---|---|---|---|---|---|
| | pyr-mg100 | $MgSiO_3$ | amorph | 2.71 | 0.2 | 500 | Dorschner+95 |
| | pyr-mg95 | $Mg_{0.95}Fe_{0.05}SiO_3$ | amorph | 2.74 | 0.2 | 500 | Dorschner+95 |
| | pyr-mg80 | $Mg_{0.8}Fe_{0.2}SiO_3$ | amorph | 2.9 | 0.2 | 500 | Dorschner+95 |
| [pyr] | pyr-mg70 | $Mg_{0.7}Fe_{0.3}SiO_3$ | amorph | 3.01 | 0.2 | 500 | Dorschner+95 |
| | pyr-mg60 | $Mg_{0.6}Fe_{0.4}SiO_3$ | amorph | 3.1 | 0.2 | 500 | Dorschner+95 |
| | pyr-mg50 | $Mg_{0.5}Fe_{0.5}SiO_3$ | amorph | 3.2 | 0.2 | 500 | Dorschner+95 |
| | pyr-mg40 | $Mg_{0.4}Fe_{0.6}SiO_3$ | amorph | 3.3 | 0.2 | 500 | Dorschner+95 |
| ens | pyr-c-mg96 | $Mg_{0.96}Fe_{0.04}SiO3$ | cryst[3] | 2.8 | **2.0** | **99** | Jäger+98 |
| ol | ol-mg50 | $MgFeSiO_4$ | amorph | 3.71 | 0.2 | 500 | Dorschner+95 |
| | ol-mg40 | $Mg_{0.8}Fe_{1.2}SiO_4$ | amorph | 3.71 | 0.2 | 500 | Dorschner+95 |
| for | ol-c-mg100 | $Mg_2SiO_4$ | cryst[3] | 3.27 | **5.0** | 200 | Suto+06 |
| | ol-c-mg95 | $Mg_{1.9}Fe_{0.1}SiO_4$ | cryst[3] | 3.33 | **2.0** | 8190 | Fabian+01 |
| fay | ol-c-mg00 | $Fe_2SiO_4$ | cryst[3] | 4.39 | **3.0** | 250 | Fabian+01 |
| | astrosil | $MgFeSiO_4$ | mixed | 3.3 | 6e-5 | 1e5 | Draine+03 |
| [c] | c-z | C | amorph? | 1.8 | 0.05 | 1e4 | Zubko+96 |
| | c-p | C | amorph | 1.8 | 0.11 | 800 | Preibisch+93 |
| gra | c-gra | C graphite | cryst[3] | 2.16? | 0.001 | 1000 | Draine+03 |
| org | c-org | CHON organics | amorph | 1.4 | 0.1 | 1e5 | Henning+96 |
| | c-nano | C nano-diamond | cryst | 2.3 | 0.02 | **110** | Mutschke+04 |
| iron | fe-c | Fe | metal | 7.87 | 0.1 | 1e5 | Henning+96 |
| [tro] | fes | FeS | metal | 4.83 | 0.1 | 1e5 | Henning+96 |
| | sic | SiC | cryst | 3.22 | 0.001 | 1000 | Laor93 |
| qua | sio2 | $SiO_2$ | amorph | 2.65 | 0.0006 | 500 | Kitamura+07 |
| cor | cor-c | $Al_2O_3$ | cryst | 4.0 | 0.5 | **40** | Koike+95 |
| [h2o] | h2o-w | Water ice | cryst | 0.92 | 0.04 | 2e6 | Warren+08 |
| | h2o-a | Water ice | amorph | 0.92 | 0.04 | 2e6 | Hudgins+93 |
| co2 | co2-w | $CO_2$ ice | cryst | 1.6 | 0.05 | 2e5 | Warren+86 |
| nh3 | nh3-m | $NH_3$ ice | cryst | 0.75 | 0.14 | 200 | Martonchik+83 |
| co | co-a | CO ice | amorph | 0.81 | **3.8** | **5.8** | Palumbo+06 |
| | co2-a / c | $CO_2$ ice | am / cr | 1.2 | **2.5** | **20** | Gerakines+20 |
| | ch4-a / c | $CH_4$ ice | am / cr | 0.47 | **2.0** | **20** | Gerakines+20 |
| | ch3oh-a / c | $CH_3OH$ ice | am / cr | 0.78/1.02 | **2.0** | **24** | Gerakines+20 |

[3] See appendix C.1 about the treatment of crystalline materials.

### 5.2. External refractory index files (`lnk` files)

`optool` can use external refractive index data in files with the following format[4]:

- The file may start with several comment lines (lines starting with `!`, `#`, or `*`).
- The next line contains two numbers, the number of wavelengths $n_\lambda$ and the specific density $\rho$ of the material in g/cm$^3$.
- The remaining lines should form three columns of data: $\lambda[\mu\text{m}]$ (sorted either up or down), and the real and imaginary parts of the refractive index, $n$ and $k$.

We provide additional data ready for use with `optool` in a separate repository. Other resources are the Jena database, ARIA and original papers in the literature. Don't forget to add the line with $n_\lambda$ and $\rho$! If that is not possible, `optool` will count the lines and you can specify the density after the mass fraction, like this: `optool -c path/to/file.lnk 0.7 3.42`. Please include references for any optical properties used in your study.

### 5.3. One-off materials

For a calculation at a single wavelength you can give the refractive index on the command line, like this: `optool 1.57:0.56:2.08 -l 0.74 -s`. This example specifies the refractive index of $m = 1.57 + 0.56i$ for a material with a density of 2.08g/cm$^3$, and the computation of the scattering matrix will be done at a wavelength of 0.74$\mu$m.

## 6. Output files

**dustkappa.dat**

This is an ASCII file containing the basic opacity results. It starts with a comment section describing the dust model and also showing the exact command line that was used to produce the file. The header is followed by the format number (3, currently), followed by the number of wavelengths in the grid, both on lines by themselves. This is followed by a block with these columns:

1. wavelength $\lambda$ [micron]
2. mass absorption cross section $\kappa_{\rm abs}$ [cm$^2$/g]
3. mass scattering cross section $\kappa_{\rm sca}$ [cm$^2$/g]
4. asymmetry parameter $g$

**dustkapscatmat.dat**

ASCII file with cross sections and full scattering matrix. It is an extended version of the `dustkappa.dat` file. This file has a format number (0), the number of wavelengths and then the number of angular points after the comment section. After an empty line, the same opacity block as in `dustkappa.dat` is present. Another empty line is followed by a list of the grid angles, another empty line, and then the scattering matrix elements for all wavelengths and all angles. The comment section at the start of the file shows the structure in a formal way. See appendix B.1 for information about the normalization of the scattering matrix and about the angular grid that is used for it. Also, see the `-radmc` switch which will modify[5] the output to make sure it can be used as an input file for RADMC-3D.

[4] This file structure is also compatible with what is needed to set the wavelength grid with **-l FILE**.

[5] This includes a change of the angular grid and a change in the normalization of the scattering matrix. The format number will be 1 (or 101 for a sparse file).

To save space, `optool` can write a *sparse file* (iformat=100) that stores the full scattering matrix only for selected wavelengths (for example, the ones that will be used for image generation). Use `-sp LAM` or `-sp LAM1 LAM2` to define a wavelength (interval)[6] for the matrix to be stored. Multiple `-sp` switches are allowed.

**dustkappa.fits**

optool itself does not write FITS files. Instead, run optool with the `-s` switch from the Python module, and write the file with the `write_fits` method:

```
import optool
p = optool.particle("optool pyr 0.8 c 0.2 -s")
p.write_fits("dustkappa.fits")
```

With `-d`, select one particle with `p.select(i)` first. This needs the `astropy` package. The FITS-file has two HDU blocks. The first contains the cross sections per unit mass (units $\mathrm{cm}^2/\mathrm{g}$). It is a $\mathrm{n}_\lambda \times 4$ matrix with these columns: wavelength in micron, $\kappa_{\mathrm{ext}}$, $\kappa_{\mathrm{abs}}$, $\kappa_{\mathrm{sca}}$. The second block is a $\mathrm{n}_\lambda \times 6 \times \mathrm{n}_{\mathrm{ang}}$ matrix, containing the 6 elements of the scattering matrix ($\mathrm{F}_{11}$, $\mathrm{F}_{12}$, $\mathrm{F}_{22}$, $\mathrm{F}_{33}$, $\mathrm{F}_{34}$, and $\mathrm{F}_{44}$) for $\mathrm{n}_{\mathrm{ang}}$ equidistant scattering angles from forward scattering (element 0) to backward scattering (element $\mathrm{n}_{\mathrm{ang}}$-1), for each $\lambda$.

**optool.tex**

As a little gimmick, you can run `optool2tex` with the exact same command line arguments as used in an `optool` run. `optool.tex` then contains text and a table, describing the methods used for the opacity computation and listing the composition of the grains. All relevant references are given - the BibTeX file `optool.bib` is required for the file to be processed properly. You can rework this text to include it into your paper. For more details, read the comment section in `optool2tex`.

**optool_mix.lnk**

When using the `-w` switch, `optool` will write the result of mixing to this file.

**optool_sd.dat & optool_lam.dat**

When using the `-w` switch, `optool` will write the grain size grid and the wavelength grid to two files. The files serve as example files for what the structure of files need to be to be read in with `-a file.dat` and `-l file.dat` for user-provided size and wavelength grids.

# 7. Python interface

`optool` comes with a `python` module[7] `optool.py` that runs `optool` in the background[8] and puts all computed quantities as `numpy` arrays into a python object. This makes it straight forward to inspect and further process the output. Here is how to use it:

```
import optool
p = optool.particle("~/bin/optool pyr 0.8 -m h2o 0.2 -na 24 -d")
```

The argument to `optool.particle()` must be a valid shell command[9] to run `optool`, if necessary with the full path to the `optool` binary. Depending on the presence of the

[6] The file will always have the matrix for at least two adjacent wavelengths around the specified $\lambda$, so that an interpolation to the exact wavelength will be stable.

[7] `optool.py` must be installed in the python environment, or be present in the current directory.

[8] The module runs the command as a subprocess, with output to a temporary subdirectory.

[9] As a string, or as a list like `["/path/to my/command","arg1","arg2",...]`.

`optool`'s **-d** switch, the command will produce opacities either for $n_p = 1$ particle, or for $n_p = n_a$ particles. Most of the attributes (with the exception of the global wavelength and angular grids) will therefore be arrays with the first dimension equal to $n_p$, even if $n_p = 1$. The resulting object will have the following attributes:

| **Attribute** | **Type/Shape** | **Quantity** |
|---|---|---|
| `cmd` | `string` | The full command given in the particle() call |
| `radmc` | `boolean` | Output follows RADMC conventions |
| `scat` | `boolean` | Scattering matrix is available |
| `nlam` | `int` | Number of wavelength points |
| `lam` | `float[nlam]` | The wavelength grid |
| `nang` | `int` | Number of scattering angles |
| `scatang` | `float[nang]` | The angular grid |
| `materials` | `[[[...]...]...  ]` | Lists with [location,$m_{frac}$,$\rho$,material] |
| `np` | `int` | Number of particles, either 1 or (with -d) $n_a$ |
| `fmax` | `float[np]` | Maximum volume fraction of vacuum for DHS |
| `pcore`, `pmantle` | `float[np]` | Porosity of the core/mantle material |
| `amin`, `amax` | `float[np]` | min/max grain size used for each particle |
| `nsub` | `int[np]` | Number of sizes averaged for each particle |
| `apow` | `float[np]` | Negative size distribution power law (e.g. 3.5) |
| `amean`, `asig` | `float[np]` | Centroid & width of (log-)normal distrbution |
| `a1`, `a2`, `a3` | `float[np]` | Mean <a>, $\sqrt{< a^2 >}$, and $\sqrt[3]{< a^3 >}$ |
| `rho` | `float[np]` | Specific density of grains |
| `kabs,ksca,kext` | `float[np,nlam]` | Absorption,scattering,extinction cross section |
| `gsca` | `float[np,nlam]` | Asymmetry parameter |
| `f11`, ..., `f44` | `float[np,nlam,nang]` | Scattering matrix element $F_{11}$, ... ,$F_{44}$ |
| `chop` | `float[np]` | Degrees chopped off forward scattering |
| `plot()` | `method` | Plot the cross sections and matrix elements |
| `computemean()` | `method` | Compute Planck/Rosseland mean opacities |
| `tmin,tmax,ntemp` | `float,float,int` | Temperature grid for mean opacities |
| `temp` | `float[ntemp]` | Temperatures used for mean opacities |
| `kplanck,kross` | `float[np,ntemp]` | Mean opacities, after calling `computemean()` |
| `norm` | `string` | Current scattering matrix normalization |
| `scatnorm()` | `method` | Check/change scat. matrix normalization |
| `sizedist()` | `method` | Sum opacities over a size distribution |

Applying the `plot()` method to a `particle` object like `p.plot()` will show (see Fig 1):

- a plot showing the opacities $\kappa_{\rm abs}$, $\kappa_{\rm sca}$, and $\kappa_{\rm ext}$ as a function of wavelength, along with the asymmetry parameter $g$ (on a linear y-scale). Note that the blue $g$ curve does not have its own axis, imagine the full $y$ axis going from 0 to 1 for $g$.
- a plot showing the scattering matrix elements as a function of scattering angle, with sliders to go through grain sizes and wavelengths. When interpreting the y axis, note that we plot the positive/negative $\log_{10}$ of positive/negative matrix elements, compressing the range from $10^{-2}$ to $10^2$ into a line (use the grey lines as a guide, ignore the y-axis labels). If you cannot see $F_{11}$, it is because it is equal to and hidden behind $F_{22}$. If you cannot see $F_{33}$, it is because it is equal to and hidden behind $F_{44}$.
- If the `computemean` method has been called first, the mean opacities $\kappa_{\rm Planck}$ and $\kappa_{\rm Ross}$ are shown in a separate plot. The mean opacities are per unit of grain mass, so please apply a dust-to-gas mass ratio to obtain opacities for a gas-dust mixture.

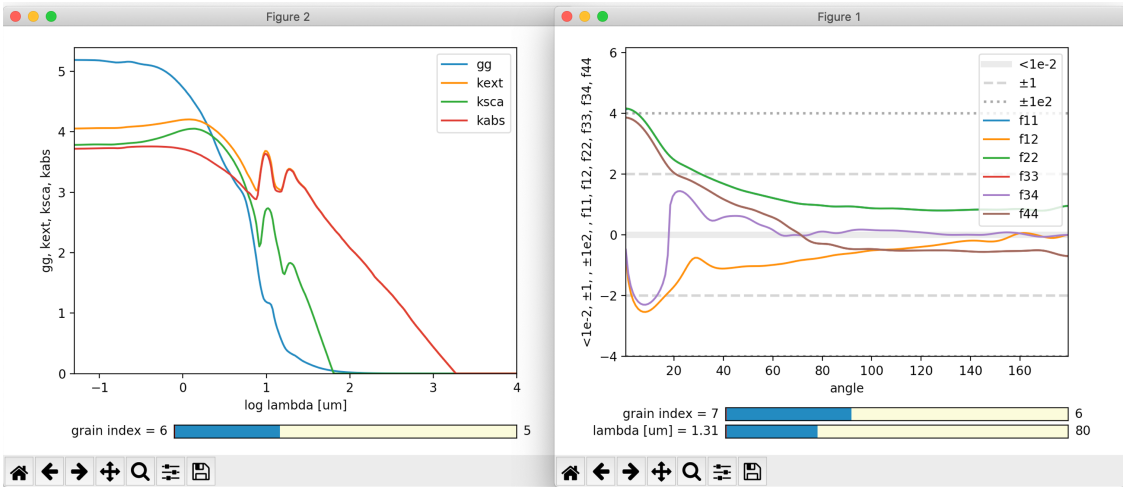



Figure 1: Plots created by running `p.plot()` on an optool particle.

The python module has a few more tricks up its sleeve (for details check the documentation inside the Python module file `optool.py`):

- You can cache results of an expensive computation for quick reloading in a new python session. When running the following command twice, `optool` will be called only the first time. The cache will reside in the specified directory.

  ```
  sil = optool.particle("optool -d ol-mg50 -na 100",cache="silcache")
  ```

- You can read the results directory created by another run of optool by leaving the command parameter empty and specifying the directory as the second parameter.

  ```
  part = optool.particle("","path/to/directory")
  ```

- A `lnktable` class to read, plot, modify and write `lnk` files.

  ```
  x = optool.lnktable("lnk_data/sio2-Kitamura2007.lnk")
  x.plot()
  ```

- Compute Planck and Rosseland mean opacities

  ```
  p = optool.particle("optool  pyr 0.87  c 0.13 -p 0.25")
  p.computemean(tmin=10.,tmax=1500.,ntemp=300)
  ```

- *Particle arithmetic*: multiplying `optool.particle` objects with factors and adding them, or applying size distributions to a pre-computed set of opacities. See the documentation of the optool module and appendix C.1 for an example.

# Appendix

## A. Size distribution

`optool` implements powerlaw, log-normal, and normal size distributions. Each of these will be subject to a minimum and a maximum grain size. The grain size grid is logarithmic, so $da \propto a$. The logarithmic bins are then filled according to:

| | |
|---|---|
| powerlaw | $n(a) \propto a^{-p+1}$ |
| log-normal distribution, triggered by $\sigma > 0$ | $n(a) \propto \exp\left[-\frac{1}{2}\left(\frac{\ln(a/a_{\rm m})}{\sigma}\right)^2\right]$ |
| normal distribution[10], triggered by $\sigma < 0$ | $n(a) \propto \exp\left[-\frac{1}{2}\left(\frac{a-a_{\rm m}}{\sigma}\right)^2\right]$ |

Other size distributions can be constructed using the python interface. Finally, `optool` can also read a size distribution from a file, and this is also the way to provide an arbitrary size grid. The first data line in the file gives the number of grain size bins, followed by lines with two numbers each: grain size in micron and number of grains in the corresponding bin. To get an example file, run `optool` with the option **`-w`**):

## B. Scattering Matrix: The fine print

### B.1. Phase function normalization

A number of different normalizations for the scattering matrix are being used in the literature and in computational tools. The differences are significant, and it is important to be aware of the choice. For `optool` we are using a convention (Hovenier (2004)) in which the average over all directions of the 1-1 element of the scattering matrix equals unity, i.e. the integral will be $4\pi$:

$$\oint_{(4\pi)} F_{11}(\lambda,\Theta)d\Omega = 2\pi\int_{-1}^{1} F_{11}(\lambda,\mu)\mathrm{d}\mu = 4\pi \quad , \tag{1}$$

with $\mu = \cos\Theta$. `optool` can also produce output for RADMC-3D which uses instead

$$\oint_{(4\pi)} Z_{11}(\lambda,\Theta)d\Omega = 2\pi\int_{-1}^{1} Z_{11}(\lambda,\mu)\mathrm{d}\mu = \kappa_{\rm sca}(\lambda) \quad . \tag{2}$$

The books by Bohren & Huffman and by Mishchenko use different normalizations again. You can change the normalization of the scattering matrix in the python interface with the `scatnorm()` method. By default, that method checks the current normalization. Using an argument `'r'`, `'b'`, `'m'`, or `'h'` will modify the normalization.

### B.2. Forward-scattering peak

Particles that are much larger than the wavelength of the considered radiation can show extreme forward scattering, where much of the *scattered* radiation is sent into just a few degrees around the forward direction. This can be difficult to handle for radiative

[10] A normal distribution is not sampled symmetrically on a logarithmic size grid - please make sure your sampling is fine enough around the mean size.

transfer codes which have limited angular resolution or limited sampling. MCMax3D has the `nspike` keyword to deal with this issue. Other tools (e.g. RADMC-3D) require this to be taken care of by the process that creates the opacity files. The `-chop` switch specifies a number of degrees around the forward scattering direction. Inside that cone, the scattering matrix gets limited to the value at the edge of the cone. To compensate and ensure energy conservation, the scattering cross section will be reduced accordingly. As a result, the radiation that would be *scattered* into this narrow range of angles will be treated as if it did have *no interaction at all* with the grain.

### B.3. Angular grid

`optool` uses an angular grid in one-degree steps from 0 to 180 degrees. The full degrees are the cell *interfaces* of that grid. `optool` computes the scattering matrix at the cell *midpoints*, i.e. at 0.5°, 1.5° etc to 179.5°, for a total of 180 values. The scattering matrix is normalized in this way, so that a numerical integral gives the correct result.
RADMC-3D requires the values of the scattering matrix on the cell *boundaries*, so at 0°, 1° etc to 180°, for a total of 181 values. For the input files for RADMC-3D, we interpolate and extend the computed values to the cell boundaries.

## C. More on optical properties

### C.1. Crystalline materials

Crystalline materials have optical properties that are dependent on the relative orientation of the electric field of the incoming radiation and the crystallographic axes of the material. A fully correct treatment would require the use of the refractive index *tensor* which is not implemented in optool (DDA can in principle do that). There are two approximations that can be used. For the materials built into `optool` we have assumed that the material consists of *many small crystalline areas that are randomly oriented within each grain*. For this, the refractive index data for different orientations have been combined using the Bruggeman effective medium rule. It results in a single refractive index data set. However, we can also think of a situation where *each dust grain is a single crystal in a cloud of randomly oriented grains*. In that case, we need to compute opacities for individual orientations, and then average the opacities. Axis-specific data is (when available) also included in the optool distribution, in the `lnk_data/ad` directory. You could use the python interface to do this kind of mixing in the following way:

```
import optool
px = optool.particle("optool -a 1.5 lnk_data/ad/c-gra-x-Draine2003.lnk")
py = optool.particle("optool -a 1.5 lnk_data/ad/c-gra-y-Draine2003.lnk")
pz = optool.particle("optool -a 1.5 lnk_data/ad/c-gra-z-Draine2003.lnk")
pmix = (px+py+pz)/3.
```

### C.2. How to ingest refractive index data for another material

Using external refractive index data means that you have to keep track of where those files are. It can be convenient to compile your favorite materials into `optool`, so that accessing them will be as simple as using the built-in materials. Here is how to do that:

1. Give your `lnk` file a name exactly like `pyr-mg70-Dorschner1995.lnk`, where the start of the name (`pyr-mg70`) is the key to access the material and `Dorschner1995` (the text after the final `-`) is the reference.

2. Put this file into the `lnk_data` directory.
3. Optionally edit `lnk_data/lnk-help.txt`, so that `optool -c` will list the new material. Note that, in order to define *generic keys*, optool looks for pairs that look like `genkey -> fullkey` in this file.
4. Run `make ingest` to update `optool_refind.f90`, now with your new material.
5. Recompile and install the code.

## C.3. Overview of optical properties

The grid plot in Fig. 2 shows the imaginary parts of all built-in materials, in the wavelength range from 0.05 to 300 $\mu$m. Some if the ices have only data in a small range, where the vibrational transitions lie. However, these materials can be used over a much broader wavelength range, because the extrapolation becomes problematic only in the UV where electronic transitions kick in.

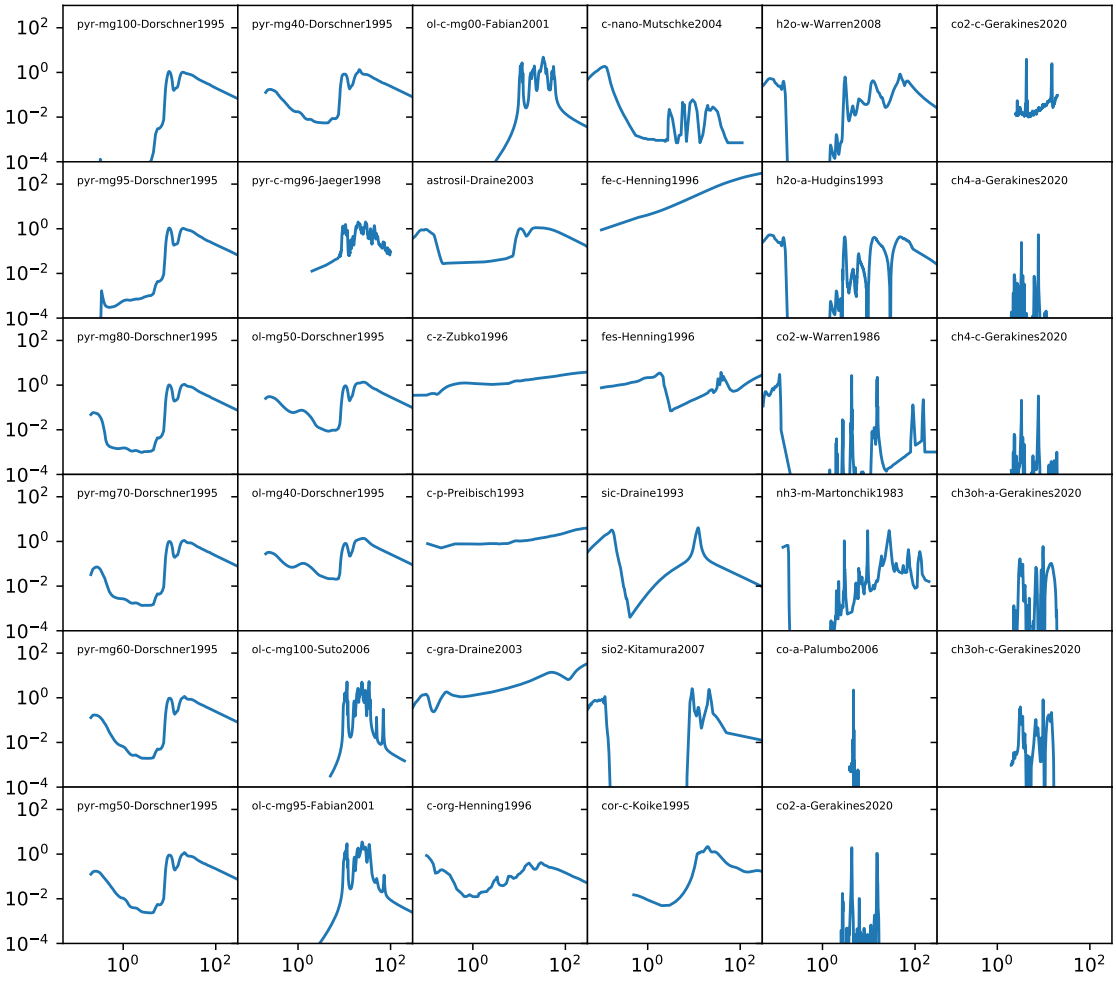



Figure 2: Imaginary part of refractive index of all builtin materials.

# D. Internals

This appendix describes some key aspects of the internal workings of the code.

### Refractive index data

Measured refractive index data is obtained from data compiled into the code, or read-in from a file. That data is then interpolated and extrapolated onto the wavelength grid requested for the computation. Extrapolation toward short wavelengths is done keeping the refractive indices constant. Extrapolation toward long wavelengths assumes that the last two measured data points define a powerlaw. Interpolation in the measured grid is done using double-logarithmic interpolation.

### Mixing

Once the refractive index for all involved materials is available, the core and the mantle mixtures are created independently, using the Bruggeman rule. Mass fractions are converted into volume fractions, and porosity is implemented using vacuum as an additional material. The subroutine doing the mixing uses an iterative procedure that is very stable, also for a large number of components.

If there is a mantle, the Maxwell-Garnett rule is applied with the core being treated as an inclusion inside a mantle matrix.

With a `-w` switch, `optool` will write result of mixing into the file `optool_mix.lnk`.

### DHS

In order to simulate irregularities in grains (irregular shapes, or the properties of low-porosity aggregates), `optool` averages the opacities of grains with an inner empty region, over a range of volume fractions of this inner region between 0 and $f_{\rm max}$. The subroutine used to compute the opacities and scattering matrix elements for these structures is `DMiLay` (Toon & Ackerman 1981). For speed, you can use `-xlim 1e3` or so to set a limit for the size parameter ($x = 2\pi a/\lambda$) where `optool` switches from DHS to Mie[11]. In rare cases, `DMiLay` might not converge and return an error. `optool` then falls back to a Mie computation, using the routine `MeerhoffMie` (de Rooij+ 1984), with a size parameter limit to ensure stability. In a situation where the imaginary part of the refractive index is extremely small, numerical inaccuracies may lead to an unphysical result with $q_{\rm sca} > q_{\rm ext}$, implying a small negative $q_{\rm abs}$. `optool` therefore enforces $q_{\rm abs} > q_{\rm ext}/10^4$.

### MMF

To construct fluffy/fractal aggregates, `optool` needs the number of monomers $N$, the fractal dimension $D_{\rm f}$, and a scaling factor $k_{\rm f}$ which are related to the radius of gyration $R_{\rm g}$ of the aggregate by $N = k_{\rm f}(R_{\rm g}/a_0)^{D_{\rm f}}$. The size $a$ of the particles as specified by the **-a** switch is interpreted as the *compact*[12] size of all material in the aggregate, so that $N = a^3/a_0^3$, where $a_0$ is the monomer radius. The average volume filling factor $f$ can be expressed by $f = N \cdot (\sqrt{3/5}\, a_0/R_{\rm g})^3$. To determine the structure of the aggregates, the user can specify a structure parameter. If that parameter is >1, it is interpreted as the *fractal dimension* $D_{\rm f}$. Using a fixed fractal dimension means that the volume filling factor will decrease with aggregate size.

[11] The default limit is $x_{\rm lim}=10^8$. A smaller value is OK in applications where the short wavelength opacity is dominated by smaller particles. `optool` does the alternative Mie computation with a slightly increased grain diameter, representing the mean geometric cross section of the DHS spheres, to connect well to the DHS opacities.

[12] still including the porosity specified with the **-p** switch (which is porosity residing in the monomers themselves), but not any "porosity" resulting from the aggregate structure

If the parameter is <1, it is interpreted as a fixed *volume filling factor* $f$ that applies to all aggregate sizes - with the implication that then the fractal dimension increases as a function of size. The fractal prefactor $k_\mathrm{f}$ is chosen automatically so that the asymptotic density of small aggregates is the monomer material density. To force another value for the prefactor, it can be given explicitly as the third value of the `mmf` option. The following table summarizes the relevant equations.

| | `-mmf A0 DF` | `-mmf A0 FILL` | `-mmf A0 DF KF` |
|---|---|---|---|
| $f$ | $N^{(D_\mathrm{f}-3)/3}$ | **given by user** | $\sqrt{27/125}\,k_\mathrm{f}^{3/D_\mathrm{f}} N^{3-1/D_\mathrm{f}}$ |
| $D_\mathrm{f}$ | **given by user** | $3\ln N \,/\, \ln(N/f)$ | **given by user** |
| $k_\mathrm{f}$ | $(5/3)^{D_\mathrm{f}/2}$ | $(5/3)^{D_\mathrm{f}/2}$ | **given by user** |

With the structure defined, `optool` then applies the formalism from Tazaki & Tanaka (2018) and Tazaki (2021) to compute cross sections and the scattering matrix. `optool` also computes the phase shift $\Delta\phi$ to check the validity of the scattering matrix. If the condition $\Delta\phi < 1$ for accurate scattering matrix results is violated, a warning will be issued and the scattering matrix will be set to zero at the relevant wavelengths. However, the opacities will remain applicable. You can request to get the result computed under the assumption of single scattering at wavelengths where the phase shift is too large. This may be usable for absorbing materials, but we do not have a clear criterion on when it will be accurate. For this result, use **-mmfss** instead of **-mmf**. `optool` will then also print the wavelength below which the scattering matrix needs to be used with caution.

### CDE

CDE (Continuous Distribution of Ellipsoids) is an analytical formalism by Bohren & Huffman (1998) to compute the opacity of a very broad shape distribution. This method is only applicable in the Rayleigh limit $x = 2\pi a \ll \lambda$ and $|mx| \ll 1$. `optool` will issue a warning if the computation leaves the bounds of this condition. The scattering matrix will be computed from a single sphere in the Rayleigh limit.

## E. Troubleshooting

1. If you get a compilation error about the intrinsic module `ISO_FORTRAN_ENV`, compile with `make oldio=true`.
2. If you get oscillations in the opacities, in particular at long wavelengths, the grain size resolution is not sufficient. Use more grain sizes (**-a**, **-na** and **-d** switches).
3. If you do not remember how to reproduce a specific run, check the output file header. It contains the exact command that was used to produce the file.
4. If the `optool` command is not found by your shell, make sure the `optool` executable is on your binary search path. Or run it by giving the full path, like `./optool`.

## F. Acknowledgments

We are indebted to

- the Jena Database of Optical Constants and the Aerosol Refractive Index Archive for their invaluable collections of refractive index datasets.
- Rens Waters, Thomas Henning, Xander Tielens, Elisabetta Palumbo, Laurent Pilon, Jeroen Bouwman, and Melissa McClure for discussions around optical properties of cosmic dust analogues.
- Charlène Lefèvre for SIGMA, which inspired me to add grain mantles.
- Kees Dullemond for discussions about the RADMC-3D input format and the scattering matrix, for the idea to write `optool2tex` and for letting me include his incredible python plotting routine `viewarr` (available on github).
- Gabriel-Dominique Marleau for testing and feedback, in particular on `optool2tex`.
- Thiébaut Schirmer for triggering the addition of a log-normal size distribution.

## G. Bibliography

- Birnstiel, T. et al. 2018, ApJ 869, 45
- Bohren, C.F. and Huffman, D.R. 1998, Wiley-VCH, *Absorption and Scattering of Light by Small Particles*
- Draine, B. 2003, ApJ 598, 1017
- Draine, B. 2003, ApJ 598, 1026
- Dorschner, J. et al. 1995, A&A 300, 503
- Fabian, D. et al. 2001, A&A 378, 228
- Gerakines, P. and Hudson, R. 2020, ApJ 901, 52
- Henning, Th. and Stognienko, R. 1996, A&A 311,291
- Hovenier, J., 2004, Report available on ADS.
- Jäger, C. et al. 1998, A&A 339, 904
- Kitamura, R. et al. 2007, Applied Optics 46,33, p. 8188
- Koike, C. et al. 1995, Icarus 114, 203
- Laor, A. and Draine, B. 1993, ApJ 402, 441
- Martonchik, J. 1984, Applied Optics 23, 541
- Min, M. et al. 2005, A&A, 432, 909
- Min, M. et al. 2016, A&A, 585, 13
- Mishchenko, M. et al. 2002, Cambridge University Press, *Scattering, absorption, and emission of light by small particles*
- Mutschke, H. et al. 2004, A&A 423, 983
- Okuzumi, S. et al. 2009, ApJ 707, 1247
- Palumbo, E. et al. 2006, PCCP 8, 279
- Preibisch, Th. et al. 1993, A&A 279, 577
- de Rooij W. and van der Stap, C. 1984, A&A 131, 237
- Steyer, T. 1974, PhD Thesis, The University of Arizona
- Tazaki, R. et al. 2016, ApJ 823, 70
- Tazaki, R. & Tanaka, H. 2018, ApJ 860,79
- Tazaki, R. 2021, MNRAS 504, 2811
- Toon, O. & Ackerman,T. 1981, Applied Optics 20, 3657
- Warren, S. and Brandt, R. 2008, JGRD,113, D14220
- Warren, S. 1986, Applied Optics 25, 2650
- Woitke, P. et al. 2016, A&A 586, 103
- Zubko, V. et al. 1996, MNRAS 282, 1321